

# Foundations of Programming

## Assignment No. : 01

### 1. **PROBLEM STATEMENT:**

Write a program in C to check whether a given year is a Leap Year

### **OBJECTIVE:**

1. To understand the use of conditional statements in C programming.
2. To implement logical conditions for solving real-world problems.
3. To develop problem-solving skills using decision control structures.

### **THEORY:**

## 1. If, if-else and if-else ladder

### 1. if Statement

#### **Theory:**

The `if` statement is used to execute a block of code only when a specified condition evaluates to true. If the condition is false, the block is skipped.

#### **Syntax:**

```
if (condition)
{
    statement;
}
```

#### **Example:**

```
if (a > b)
{
    printf("a is greater");
}
```

---

### 2. if-else Statement

#### **Theory:**

The `if-else` statement executes one block of code if the condition is true and another block if the condition is false, ensuring one of the two blocks always runs.

#### **Syntax:**

```
if (condition)
{
    statement1;
}
else
{
    statement2;
}
```

### **Example:**

```
if (a > b)
{
    printf("a is greater");
}
else
{
    printf("b is greater");
}
```

---

## **3. if-else Ladder**

### **Theory:**

The `if-else ladder` is used to check multiple conditions in sequence. When a true condition is found, its block is executed and the remaining conditions are skipped.

### **Syntax:**

```
if (condition1)
{
    statement1;
}
else if (condition2)
{
    statement2;
}
else
{
    statement3;
}
```

### **Example:**

```
if (marks >= 75)
{
    printf("Distinction");
}
else if (marks >= 60)
{
    printf("First Class");
}
else
{
    printf("Pass");
}
```

## 2. Leap Year

A leap year is a year that contains **366 days** instead of the usual 365 days. In a leap year, an extra day (29th February) is added to the calendar. This adjustment is necessary because the Earth takes approximately **365.25 days** to complete one revolution around the Sun. Without adding the extra day, the calendar would slowly shift away from the actual seasons.

### Rules for Leap Year:

- A year is considered a leap year if it is divisible by **400**.
- A year is also a leap year if it is divisible by **4** but **not divisible by 100**.
- Years divisible by 100 but not by 400 are **not** leap years.

These rules ensure long-term accuracy of the Gregorian calendar and prevent seasonal drift.

## 3. Algorithm:

1. Start
2. Declare an integer variable year
3. Display a message to enter a year
4. Accept the year from the user
5. Check if the entered year is greater than or equal to 0
6. If the year is divisible by 400 **OR** divisible by 4 but not divisible by 100, then
  - a. Display "Given year is a leap year"
7. Else
  - a. Display "Given year is NOT a leap year"
8. If the entered year is less than 0, display "Invalid Input"
9. Stop

## 4.Code:

```
#include<stdio.h>

int main(){

    int year;

    printf("Enter any year \n");

    scanf("%d",&year);

    if (year>=0){

        if ((year%4==0 && year%100!=0)||(year%400==0)){

            printf("Given year %d is a leap year",year);

        }

        else{

            printf("Given year %d is NOT a leap year",year);

        }

    }

    else{

        printf("Invalid Input");

    }

}
```

```

1  #include<stdio.h>
2
3  int main(){
4      int year;
5      printf("Enter any year \n");
6      scanf("%d",&year);
7
8      if (year>=0){
9          if ((year%4==0 && year%100!=0)|| (year%400==0)){
10             printf("Given year %d is a leap year",year);
11         }
12         else{
13             printf("Given year %d is NOT a leap year",year);
14         }
15     }
16
17     else{
18         printf("Invalid Input");
19     }
20
21 }
22
23
24 }

```

**PLATFORM:** Linux

**INPUT:** Enter a year (integer value).

```

Enter any year
2028|

```

**OUTPUT:** Displays whether the entered year is a Leap Year or Not a Leap Year.

```

Given year 2028 is a leap year
PS C:\Users\Manish\OneDrive\Desktop>

```

**Sample Input**

2024

**Sample Output**

2024 is a Leap Year

**CONCLUSION:** Thus, the program successfully checks whether a given year is a Leap Year using conditional statements in C.

**FAQs:**

1. What is a leap year?
2. Why is the year 1900 not a leap year but 2000 is?
3. Which operators are used in this program?
4. Difference between if and if-else statements.